

1. In optical mouse sensors, what function does the Image Processing Engine (DSP) perform on consecutive frame snapshots?

- (A) Digital watermark filtering
- (B) Optical surface cross-correlation analysis
- (C) Analog refresh rate synchronization
- (D) Interpolation of RGB values

2. Why do optical gaming mice use Infrared LEDs or Lasers instead of standard visible Red LEDs?

- (A) To reduce overall power consumption by half
- (B) To illuminate surface micro-textures with higher contrast
- (C) To transmit wireless positioning data directly
- (D) To bypass legal laser safety limits

3. What technique is implemented in keyboard firmware to prevent mechanical switch physical vibrations from causing multiple registered inputs?

- (A) Debouncing
- (B) Multiplexing
- (C) Key Rollover
- (D) Scan-code Translation

4. In Projected Capacitive Touchscreens (Pcap), how is multi-touch location detected simultaneously?

- (A) By measuring change in mutual capacitance at grid intersections
- (B) By calculating the voltage drop across top and bottom PET layers
- (C) By timing acoustic wave reflection delays
- (D) By tracking shadow patterns cast on corner cameras

5. Which optical scan technology uses an oscillating laser beam combined with a rotating polygonal mirror to read 1D barcodes?

- (A) CCD Array Scanner
- (B) Laser Barcode Reader
- (C) Area Imager
- (D) Contact Image Sensor

6. What is the role of Diodes installed in a mechanical keyboard switch matrix circuit?

- (A) Powering individual key RGB illumination
- (B) Preventing current feedback loops that cause ghosting
- (C) Accelerating electrical polling frequency
- (D) Smoothing voltage spikes during key press

7. What parameter distinguishes a Laser Mouse from a traditional Optical LED mouse?

- (A) Laser light penetrates surface grooves for tracking on glass
- (B) Laser mice do not require a digital signal processor
- (C) Laser mice use mechanical rollers for Y-axis motion
- (D) Laser mice operate only at lower DPI levels

8. In magnetic card readers (e.g., credit card magnetic stripe readers), data is encoded on magnetic tracks using which encoding method?

- (A) NRZ-I (Non-Return-to-Zero Inverted)
- (B) F2F (Aiken Biphase / Two-Frequency Coherent Phase)
- (C) Manchester Encoding
- (D) 8b/10b Encoding

9. What advantage does a Surface Acoustic Wave (SAW) touchscreen offer over a resistive touchscreen?

- (A) 100% glass clarity without conductive coatings
- (B) Works perfectly even if covered in heavy dust or oil
- (C) Consumes less battery power than resistive screens
- (D) Supports soft rubber stylus input exclusively

10. In modern Optical Character Recognition (OCR), what is the purpose of the Binarization preprocessing step?

- (A) Translating text to ASCII binary output
- (B) Converting grayscale images into pure black and white pixels
- (C) Resizing image dimensions to standard A4 size
- (D) Correcting skewed scan angles

11. What is the operational mechanism of a Hall Effect Key Switch in advanced gaming keyboards?

- (A) Optical light beam interruption
- (B) Magnetic sensor detecting movement of a magnet stem
- (C) Physical contact between gold-plated leaf springs
- (D) Piezoelectric pressure generation

12. Which metric measures the maximum acceleration (in Gs) a mouse sensor can handle before losing tracking accuracy during fast flicks?

- (A) IPS Rate
- (B) Max Acceleration Rating (G)
- (C) Polling Jitter Rate
- (D) Lift-Off Distance

13. What function does Lift-Off Distance (LOD) serve in optical mouse customization?

- (A) Height above tracking surface where sensor stops tracking movement
- (B) Minimum physical distance required for click actuation
- (C) Height offset of mouse feet pads
- (D) Distance wireless signals travel to USB receiver

14. What type of microphone transducer relies on changes in electrical capacitance between a flexible diaphragm and a fixed backplate?

- (A) Dynamic Microphone
- (B) Condenser Microphone
- (C) Ribbon Microphone
- (D) Carbon Microphone

15. Which scanning parameter determines the range of dark to light tones a document scanner can distinguish?

- (A) Optical Resolution
- (B) Optical Density (Dmax)
- (C) Interpolated Resolution
- (D) Scan Rate (PPM)

16. In graphic design digitizers, what parameter measures tilt sensitivity of the stylus pen?

- (A) Azimuth and Altitude angle degrees
- (B) Pressure sensitivity levels (e.g., 8192)
- (C) LPI density
- (D) Report frequency in Hz

17. In linear barcodes (e.g., UPC-A), what is the function of the Guard Patterns located at the left, center, and right edges?

- (A) Store manufacturer region codes
- (B) Provide synchronization baseline and scanning boundaries
- (C) Contain mathematical checksum digits
- (D) Encryption keys for product security

18. Why are Drum Scanners used primarily in high-end archival printing rather than flatbed scanners?

- (A) They use Photomultiplier Tubes (PMT) offering superior dynamic range and resolution
- (B) They scan documents faster using LED bars
- (C) They capture 3D volumetric images automatically
- (D) They do not require optical lenses

19. How does an Optomechanical Keyboard Switch differ from a traditional mechanical switch?

- (A) Actuation occurs when a stem blocks or unblocks an infrared light beam
 (B) Actuation depends on closing an iron leaf contact
 (C) It uses liquid crystal contacts under every key
 (D) It converts key movement into RF radio signals

20. In digital cameras and webcams, what artifact occurs when shooting fast movement due to sequential row reading in rolling shutter CMOS sensors?

- (A) Jello / Skew artifact
 (B) Chromatic aberration
 (C) Barrel distortion
 (D) Moire fringing

21. What is the purpose of Angle Snapping (Grid Tracking) feature in optical mouse driver software?

- (A) Smoothing cursor trajectories into perfectly straight horizontal or vertical lines
 (B) Automatically adjusting DPI based on cursor speed
 (C) Preventing hardware acceleration curves
 (D) Restricting cursor movement inside active window borders

22. In OCR software algorithms, what is the key difference between Matrix Matching and Feature Extraction?

- (A) Feature Extraction analyzes structural line traits (loops, strokes, intersections)
 (B) Matrix Matching operates on vector PDF files only
 (C) Feature Extraction compares pixel-by-pixel against fixed templates
 (D) Matrix Matching is invariant to font size and rotation

23. What component in a biometric fingerprint scanner uses a prism and light refraction to capture fingerprint ridge contact patterns optically?

- (A) Total Internal Reflection (TIR) optical assembly
 (B) Ultrasonic transducer array
 (C) Thermal semiconductor sensor
 (D) Capacitive silicon matrix

24. High-end webcams utilize Time-of-Flight (ToF) sensors alongside RGB camera lenses to perform:

- (A) Real-time 3D depth mapping and fast laser autofocus
 (B) Optical zoom focal adjustments
 (C) Infrared night vision illumination
 (D) Frame rate upscaling to 120 FPS

25. In modern USB Human Interface Device (HID) protocol, how do input devices report data to the host operating system?

- (A) Via periodic USB Interrupt In Transfers based on polling interval
 (B) Via continuous USB Isochronous streaming
 (C) Via hardware IRQ line interrupts bypassing CPU
 (D) Via Direct Memory Access (DMA) channel requests

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(B)	6	(B)	11	(B)	16	(A)	21	(A)
2	(B)	7	(A)	12	(B)	17	(B)	22	(A)
3	(A)	8	(B)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(B)	19	(A)	24	(A)
5	(B)	10	(B)	15	(B)	20	(A)	25	(A)